

ENGR 4321 Microcontroller and Embedded Systems

[Wokwi] Simulation Assignment 3: ESP32 WiFi, APIs, and Cloud Data

Report Due Date: Monday, February 23, 2026

The objective of this assignment is to introduce ESP32-based wireless communication using WiFi, cloud APIs, and JSON-formatted data through simulation. Students will design, simulate, and document an Internet-connected embedded system using the Wokwi ESP32 simulator, reinforcing concepts related to networking, data exchange, and system-level design.

Assignment Description: This assignment emphasizes practical WiFi connectivity on embedded systems, client-server interaction using web APIs, reading and parsing JSON data, professional documentation and version control using GitHub.

Part 1. Part 1: ESP32 WiFi + API Simulation

Each student will select one **IoT-style project** inspired by the SunFounder ESP32 Starter Kit IoT examples (e.g., Adafruit IO, sensor dashboards, cloud data logging) and recreate or adapt it in Wokwi. Here is the link to the projects: [SunFounder ESP32 Starter Kit Documentation](#)

Your project must demonstrate:

- ESP32 connecting to WiFi
- Communication with a web service or API
- Data exchange using JSON

You may:

- Push sensor data to a cloud service
- Retrieve data from a public API (weather, time, IoT feed, etc.)
- Simulate a basic IoT monitoring or control system

Technical Requirements

1. **Microcontroller:** Wokwi ESP32
2. **Networking:** WiFi connection using (**WiFi.h**), **HTTP** or **HTTPS** client communication
 - a. **Important Note:** Use the special network Wokwi-GUEST with no password
3. **Data Format:** Use **JSON** for sending or receiving data, and demonstrate parsing or constructing JSON in code
4. **Input / Output (Minimum One Each)**
 - a. **Input:** Sensor, button, or simulated data source
 - b. **Output:** Serial Monitor, LED, display, or actuator

Part 2: GitHub Repository: Create a **public GitHub repository** named: **ENGR4321-ESP32-WiFi-Assignment**. Your repository must include:

1. ESP32 Arduino source code (.ino)
2. A professional README.md that includes:
 - Project overview
 - System block diagram
 - Description of WiFi and API interaction
 - Screenshot or link to Wokwi simulation

Deliverables

Each student must submit a **Simulation Assignment Report (SAR 3)** to the Canvas portal. The report must be in **Microsoft Word (.docx)** format and make use of the template available on Canvas.

Important Notes:

1. **This assignment is an individual work.**
2. Submit your report by uploading it to the Canvas **SAR 3 Portal**.